

Skills Summary

Explaining Legal Equation Transformations

Use this reference while completing your worksheet or when studying.

Skill 1 — Naming the property behind a step

Rule: A step is *legal* when the new equation has exactly the same solutions as the old one. Two families do that:

- do the same thing to **both sides** — Addition, Subtraction, Multiplication, or Division Property of Equality (the multiplier or divisor must be nonzero);
- rewrite **one side** into an equal form — Distributive Property, combining like terms.

Example: Solve $2x - 5 = 11$ and justify.

Step 1. The x is multiplied then has 5 subtracted, so I undo in reverse order: the subtraction first, then the multiplication.

Step 2. Add 5 to both sides (Addition Property): $2x = 16$. Divide both sides by 2, nonzero (Division Property). $\Rightarrow x = 8$

Try it: Solve $5x + 4 = 24$, naming the property on each line.

check: $x = 4$

Skill 2 — Deciding whether two equations match

Rule: Rewriting one side is legal only when the two forms are equal for **every** value of the variable. Distribute to *all* terms inside the parentheses; a partial distribution is a different expression, so the equation it sits in is a different equation.

Example: Write $3(x - 4)$ without parentheses.

Step 1. Nothing is being done to both sides here — I am only rewriting one side, so I need an identity, and the Distributive Property is the one that removes parentheses.

Step 2. Multiply the 3 by *both* terms: $3 \cdot x$ and $3 \cdot (-4)$. $\Rightarrow 3(x - 4) = 3x - 12$, equal for every x

Try it: Write $5(x + 3)$ without parentheses.

check: $5x + 15$

Skill 3 — Finding the step that is not legal

Rule: Read down someone's work and ask of each line: *what was done, and was it done to both sides in full?* The three usual offenders are a change made to one side only, a partial distribution, and a division by a quantity that could be zero. Then substitute the answer into the **original** equation — a legal chain always survives that test.

Example: A student writes $4x + 9 = 25 \rightarrow 4x = 34$. Find the bad line and fix it.

Step 1. The 9 moved but got bigger, so the student added 9 to the right while removing it from the left — not the same operation on both sides.

Step 2. Subtract 9 from both sides: $4x = 16$, then divide both sides by 4. $\Rightarrow x = 4$

Try it: A student writes $7x - 2 = 19 \rightarrow 7x = 17$. What went wrong, and what is x ?

check: $x = 4$

Skill 4 — Multiplying an inequality by a negative

Rule: Every equality property has an inequality twin, with one change: multiplying or dividing both sides by a **negative** number *reverses* the symbol, because negating reflects the number line and reverses order. Adding or subtracting never does.

Example: Solve $-4x < 12$.

Step 1. The only step needed is dividing both sides by -4 , and that divisor is negative, so the symbol must turn around on that line.

Step 2. $\frac{-4x}{-4} = x$ and $\frac{12}{-4} = -3$, with $<$ becoming $>$. $\Rightarrow x > -3$

Try it: Solve $-5x > 20$.

check: $x > -4$